

Nora Hernández

Full name: Nora Patricia HERNÁNDEZ LÓPEZ

PhD in AI Education

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Research interests

AI (in) education; Culturally relevant & Cross-cultural Education; Learning sciences; STEAM education; Learning analytics; Bayesian Methods

Education

- 2021 – 2026 **The University of Hong Kong** – Hong Kong
Ph.D., Education
Supervisor: Dr. Xiao Hu; Co-supervisors: Dr. Samuel K. W. Chu, Dr. Gary K. W. Wong
- 2018 – 2020 **Tecnologico de Monterrey** – Mexico
M.Sc., Computer Science
Supervisor: Dr. Jorge Adolfo Ramírez Uresti
GPA: 97/100.
Universidad del Valle de Mexico – Mexico
M.Ed., Innovation and Educational Technology
GPA: 9.5/10.
- 2011 – 2016 **Tecnologico de Monterrey** – Mexico
B.Eng., Mechatronics
GPA: 92/100.

Teaching Qualifications

- 2022 **Certificate in Teaching and Learning in Higher Education** – The University of Hong Kong
Focused on: course design, active learning, and assessment for learning.

Teaching Experience

- 2023 – 2024 **Teaching Assistant** – The University of Hong Kong
Course: Information system analysis and development (Master's level)
Led special sessions on database design; provided individual consultations; graded and provided feedback on students' assignments.
- 2021 **Private Online Tutor (Computer Science & Mathematics)**
Provided one-to-one tutoring to students aged 8–14, breaking down complex concepts into intuitive explanations; tailored instruction to individual learning styles, following international curriculum. Maintained regular feedback sessions with parents.

2016 – 2020 **Curriculum Designer** – Aerobot Planet, Mexico
Co-created a summer-camp and extra-curricular curriculum for robotics and coding courses for 8–14 year olds, incorporating project-based learning and formative assessment.

Research Experience

- 2022 – present **Research Consultant** – Impact Analytics
- Supported various youth education projects through impact evaluation research, building comprehensive MEL frameworks including survey design, data collection, and stakeholder feedback sessions.
 - Used qualitative and quantitative learning data to measure program effectiveness, learner progress, and SROI.
 - Synthesised findings into reports for clients and created materials to disseminate project outcomes to the wider community of stakeholders.
 - Facilitated regular review meetings with project partners, ensuring alignment on objectives and timelines.
- 2021 – 2026 **Doctoral Researcher** – The University of Hong Kong
- Designed and implemented culturally relevant AI literacy curricula for 200+ secondary students in Mexico and Hong Kong.
 - Led mixed-methods evaluation using Bayesian ordinal regression, topic modeling, and semi-structured interviews.
 - Published findings in peer-reviewed conferences (AIED, ICALT, LAK).
- 2020 **Master’s Thesis Researcher** – Tecnológico de Monterrey
- Built a Contextual Hybrid Bayesian Recommender System using Bayesian networks and expert knowledge elicitation.
 - Conducted a usability study (N=97) demonstrating improved practical usefulness over state-of-the-art baselines.

Journal Articles

- 2026 **Collaborating with classmates and AI to create music pieces: A topic modeling and sentiment analysis approach**
Under review.
Ng, D.T.K., **Hernández López, N. P.**, Hu, X.
- 2025 **Interaction patterns in AI Education: A comparison of two contexts**
Under review.
Hernández López, N. P., Hu, X., Ng, D.T.K.
- 2025 **Computing Education Meets the Arts: Insights from a Systematic Review**
Under review.
Hernández López, N. P., & Hu, X.

Peer-reviewed Conference Proceedings

- 2026 **Sequence Analysis for Understanding Constructionist Learning in AI Education**
In *2026 IEEE International Conference on Advanced Learning Technologies (ICALT)*.
Hernández López, N. P., & Hu, X. *In Press*.
- 2025 **Insights from Culturally Relevant AI Education Programme for Secondary School Students**
In *26th International Conference on Artificial Intelligence in Education (AIED25)*.
Hernández López, N. P., & Hu, X., Ng, D.T.K. DOI: [10.1007/978-3-031-98465-5_35](https://doi.org/10.1007/978-3-031-98465-5_35)
- 2025 **WekiMusic: Machine Learning Music Activities to Foster Constructionist AI Education**
In *2025 IEEE International Conference on Advanced Learning Technologies (ICALT)*.
Hernández López, N. P., & Hu, X. DOI: [10.1109/ICALT64023.2025.00035](https://doi.org/10.1109/ICALT64023.2025.00035)

Full list available upon request.

Book Chapters

- 2024 **Understanding Learning in Culturally Relevant Artificial Intelligence Education**
Hernández López, N.P. In Dimitriadis, Y., Cerezo, R., Balderas, A., Martínez-Monés, Spikol, D. (Eds.), *Doctoral Consortium of the Learning Analytics Summer Institute Europe 2024 (LASI Europe 2024 DC)*. CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-3738/>
- 2019 **Razonamiento estadístico**
Estrada Real, A. & **Hernández López, N.P.** Razonamiento estadístico. (2019). In Francisco Cantú & Rocío Aldeco (Eds.), *Razonamiento Computacional* (pp.63-102). Academia Mexicana de Computación, A.C. <https://amexcomp.mx/media/publicaciones/conocimiento-y-razon-comp.pdf>

Invited Talks

- 2024 **Brown Bag Seminar: AI and its implications for Education** – The University of Hong Kong
Unit of Social Contexts and Policies for Education
- 2023 **Panel de discusión: El papel de la mujer en la ciencia** – Hong Kong
Consulate General of Mexico in Hong Kong

Conference Presentations

- 2022 **IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)** – Hong Kong
Second Runner Up: TALE 2022 Interdisciplinary Postgraduate Student Forum
The Hong Kong Polytechnic University

- 2022 **Forum for Social Experiments for AI in Education** – China
Presentation: Computing Education meets the Arts: Preliminary results from a systematic review
 Graduate School of Education, Peking University
- 2020 **Congreso de Investigación y Desarrollo (CIDTEC)** – Mexico
Poster presentation: A Context Hybrid Bayesian Model for Movie Recommendations
 Tecnológico de Monterrey

Grants & Fellowships

- 2021 – 2026 Hong Kong PhD Fellowship (HKPF) – Research Grants Council, Hong Kong
Competitive university-wide fellowship, awarded to top 0.5% of applicants
- 2021 HKU Presidential PhD Scholarship – The University of Hong Kong
- 2018 National Postgraduate Scholarship (CONACyT) – Mexico
Full funding for Master's studies
- 2018, 2011 Academic Excellence Scholarship – Tecnológico de Monterrey, Mexico

Professional Development

- 2024 Doctoral Consortium – AIED24 (Recife, Brazil) & LASI Europe (Jerez de la Frontera, Spain)
- 2023 Learning Analytics Summer Institute (LASI) – Singapore (Erik Duval Scholar)
 Bayesian Statistics for Social and Health Scientists in R and Python – Online (Institute for Statistical and Data Science)
- 2022 Learning Analytics Summer Institute (LASI) – Canada (online) (Erik Duval Scholar)

Technical Skills

Learning & Development: Curriculum and Assessment Design, LMS workflows, Microsoft Office (Advanced Word, Excel, PowerPoint).

Domain Expertise: AI Literacy, AI/ML Ethics, Learning Analytics, Educational Data Mining, Sequence Process Mining, STEAM Education.

Languages & Tools: Python (Pandas, NumPy, Scikit-learn), R (tidyverse, brms, lme4, Shiny), Git, $\text{\LaTeX}$.

Statistical Methods: Bayesian Inference (MCMC, Hierarchical Modeling), A/B Testing, Hypothesis Testing, Regression Analysis, Time Series, Causal Inference.

Languages: Spanish (native), English (fluent), German (basic).

References

Available upon request.